LifeWell_XAI: Life Satisfaction Detection System

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***Abstract*—Understanding and predicting user satisfaction is essential for improving the quality and effectiveness of digital systems and services. This paper presents *LifeWell_XAI*, a machine learning–based system designed to predict user satisfaction levels based on user-provided attributes and interaction data. The proposed approach employs supervised learning models to analyze patterns within the dataset and classify satisfaction outcomes. Multiple machine learning classifiers are trained and evaluated to identify the most effective model in terms of accuracy and reliability. The best-performing model is deployed through a web-based interface, allowing real-time satisfaction prediction. Experimental results demonstrate that the proposed system achieves strong predictive performance, indicating its potential applicability in data-driven decision-making and user experience optimization.**

***Index Terms*—Explainable AI, Disease Prediction, Machine Learning, Healthcare Systems, Symptom-Based Diagnosis**

I. Introduction

Understanding whether an individual is satisfied with their life is an important aspect of social, psychological, and data-driven research. Life satisfaction reflects a person's overall well-being and perception of their quality of life, which can be influenced by multiple personal, social, and behavioral factors. However, identifying life satisfaction levels through traditional surveys and manual analysis is often time-consuming and inefficient, especially when dealing with large-scale user data.

The problem addressed in this research is determining whether a person is satisfied with their life based on a limited set of user responses. Specifically, this study aims to predict a user's life satisfaction status using a small number of carefully selected questions, reducing the burden on users while maintaining reliable prediction performance.

To solve this problem, this research proposes a machine learning–based approach where multiple supervised learning models are trained on a life satisfaction dataset. Algorithms including Logistic Regression, K-Nearest Neighbors (KNN), Naive Bayes, Gradient Boosting, and Support Vector Machine (SVM) are employed to learn patterns from user responses. The models are evaluated using standard performance metrics, and the most effective model is selected for deployment. This approach enables automated and scalable prediction of life satisfaction outcomes.

Previous studies have addressed user satisfaction and well-being prediction using machine learning techniques. Many existing works apply models such as Logistic Regression, SVM, and ensemble-based approaches to classify satisfaction levels. Some studies further enhance their systems by incorporating explainability techniques, such as SHAP, to interpret model decisions and feature importance. These approaches demonstrate the effectiveness of machine learning in satisfaction prediction tasks.

Despite these contributions, several research gaps remain. Many existing systems rely on a relatively large number of survey questions, which can reduce user participation and practical usability. In addition, some approaches focus heavily on model explainability while overlooking simplicity, user experience, and deployment feasibility. Furthermore, not all studies explore the performance of simpler probabilistic and instance-based models within the same framework.

The novelties of this paper can be summarized as:

- **Minimal Question-Based Satisfaction Prediction:** The proposed system predicts whether a user is satisfied with their life using responses from only seven carefully selected questions, significantly reducing user effort compared to traditional long-form surveys while maintaining reliable prediction performance.
- **Multiple Machine Learning Model Integration:** The system implements and evaluates multiple supervised machine learning models, including Logistic Regression, K-Nearest Neighbors, Naïve Bayes, Gradient Boosting, Support Vector Machine, and BART, allowing comprehensive performance comparison and robust model selection.
- **Incorporation of the BART Model for Satisfaction Prediction:** In addition to traditional machine learning algorithms, this research integrates the BART (Bidirectional and Auto-Regressive Transformers) model to capture contextual and semantic patterns in user responses, enhancing the system's ability to model complex relationships within the data.
- **Inclusion of Probabilistic and Instance-Based Models:** Unlike many existing studies that focus primarily on linear or ensemble models, this work incorporates probabilistic (Naïve Bayes) and instance-based (KNN) classifiers to broaden analytical diversity and performance evaluation.
- **User-Friendly Web Interface:** A visually appealing and intuitive web-based platform is designed to allow users to easily answer questions and instantly receive satisfaction predictions, ensuring accessibility even for users with limited technical skills.
- **Dark Mode Supported Interface:** The system includes dark mode support to enhance user comfort and accessibility, particularly in low-light environments, improving overall user experience and real-world usability.

By addressing the limitations identified in previous studies, the proposed system offers a practical, lightweight, and user-centric solution for automated life satisfaction prediction.

II. Literature Review

Machine learning techniques have been widely adopted to analyze subjective human-centered outcomes such as life satisfaction, quality of life, and user satisfaction by learning patterns from structured survey data. Early studies demonstrated that supervised learning models, including Logistic Regression, Support Vector Machine (SVM), and Naïve Bayes, are effective in classification tasks involving user responses and satisfaction indicators. These approaches showed strong performance in mapping survey-based inputs to satisfaction levels and provided a foundation for data-driven satisfaction analysis [1]–[3], [12]

With the advancement of ensemble learning methods, researchers explored tree-based and boosting models to improve predictive accuracy. Random Forest and Gradient Boosting classifiers were shown to outperform traditional linear models by capturing complex and non-linear relationships among user attributes [5], [8], [10]. Chen and Guestrin introduced XGBoost, a scalable tree boosting framework that significantly improved performance and computational efficiency in structured data prediction tasks [18]. These ensemble approaches have since become popular choices for satisfaction and quality-of-life prediction.

Several studies have focused specifically on predicting life satisfaction and quality of life using machine learning models. Khan *et al.* demonstrated that supervised learning algorithms can effectively predict life satisfaction outcomes and highlighted the usefulness of explainability techniques for interpreting model behavior [16].. Similarly, Lee *et al.* applied machine learning models to predict quality of life in aging populations, showing that demographic and behavioral features play a significant role in satisfaction-related outcomes [17]. These studies confirm the applicability of machine learning for modeling subjective well-being.

Instance-based and probabilistic models have also been investigated for satisfaction prediction. K-Nearest Neighbors (KNN) has been used to classify users by measuring similarity between response patterns, showing competitive results for small to medium-sized datasets [6], [7]. Naïve Bayes classifiers have been applied due to their simplicity, fast training time, and effectiveness in handling categorical survey data [4], [6]. Despite their effectiveness, these models are often excluded from large comparative studies in favor of more complex algorithms.

Beyond individual-level attributes, external and contextual factors have been examined in satisfaction research. Studies on local environmental quality in Germany revealed that environmental conditions such as pollution levels, urban infrastructure, and green spaces have a measurable impact on life satisfaction [19]. These findings emphasize that satisfaction is influenced not only by personal factors but also by environmental and social conditions, although such variables are often omitted in lightweight predictive systems.

Recent research trends also emphasize model transparency and interpretability. Several studies integrate explainable artificial intelligence techniques, such as SHAP, to interpret feature importance and improve trust in satisfaction prediction systems [16]. While explainability enhances model understanding, many systems prioritize interpretability over simplicity and practical deployment, which can limit usability in real-world applications.

Finally, usability and system deployment have become important considerations in satisfaction prediction research. Web-based platforms have been proposed to make machine learning models accessible to end users; however, many focus primarily on offline analysis and lack attention to user interface design and accessibility [12], [13]. Features such as minimal question-based input, real-time prediction, and enhanced user experience are rarely addressed together. These limitations motivate the development of a user-centric system that combines reduced input requirements, multiple model evaluation, and practical deployment.

III. METHODOLOGY

A. System Flow

The proposed methodology follows a structured pipeline to develop an intelligent system for predicting life satisfaction using machine learning with explainable AI analytics. The complete system flow is illustrated in Fig. 1, which encompasses data collection, preprocessing, model training, evaluation, and deployment into a user-friendly interface. The process begins with defining the research goal: developing an interpretable healthcare application that utilizes symptom-based medication suggestions and potential side effects to create a user-friendly interface. Data collection involves gathering comprehensive datasets on health indicators, satisfaction metrics, and demographic factors. Exploratory data analysis (EDA) follows, including data loading, missing data analysis, and visualization to understand distributions and correlations. Data preprocessing includes cleaning, imputation of null values, and handling categorical data. The dataset is then split into training and test sets. Multiple machine learning models are trained, including Support Vector Classifier (SVC), Gradient Boosting, Naive Bayes, and others. Model selection is based on performance metrics, and the best model is saved using pickle for deployment. Post-training, the system integrates custom helper functions for predictions and evaluations. The final steps involve web interface development using Streamlit, testing, and deployment. This flow ensures scalability, interpretability, and practical usability for end-users assessing their well-being profiles.

B. Dataset Description

The dataset utilized in this study is the SHILD2012 Census English version for individuals aged 16–64, sourced from a comprehensive survey on health, life satisfaction, and socio-economic factors. It comprises 18,957 records with 384 columns capturing diverse attributes such as demographics (e.g., gender, age), health status (e.g., general health, stress

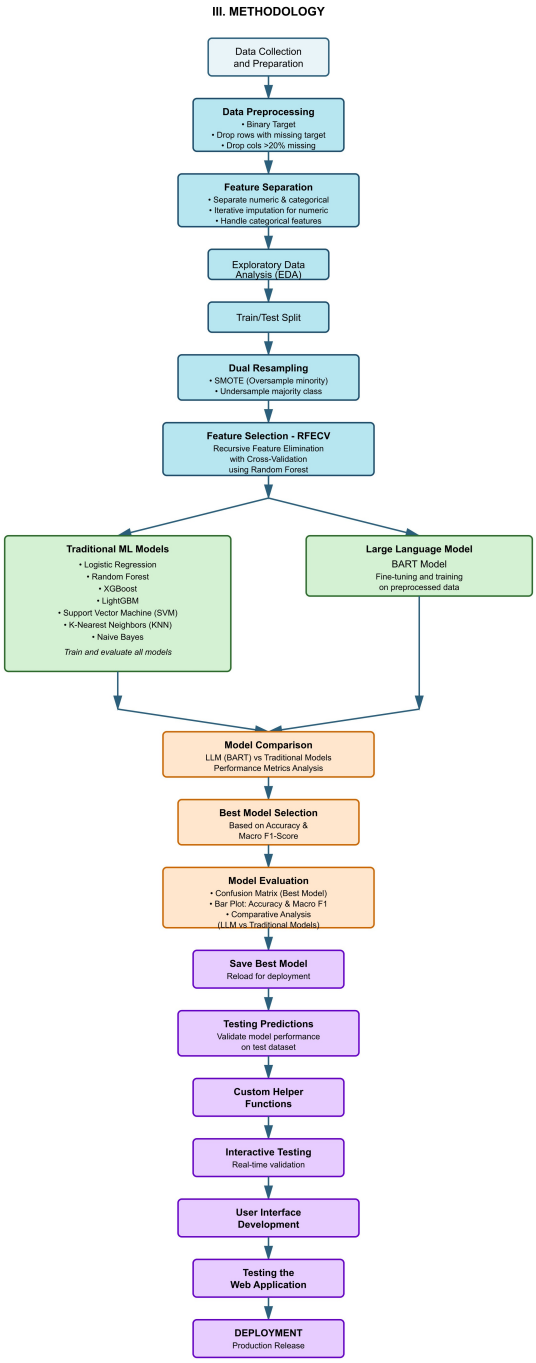


Fig. 1: Flowdiagram of Methodology

levels), satisfaction metrics (e.g., financial satisfaction, housing satisfaction), social connections (e.g., community engagement, social support), and behavioral indicators (e.g., frequency of social interactions, employment status). Key features include:

- **Demographics**: Age, gender, education level, job status.
- **Health and Well-being**: General health status, stress levels, symptom-based data.
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- **Satisfaction Indicators**: Life satisfaction scores, financial/housing contentment.
- **Social Factors**: Community engagement, social support quality.

C. Preprocessing Techniques and Models

1) Preprocessing: Data preprocessing is crucial for handling the heterogeneous nature of the SHILD2012 dataset. The steps include:

1) **Data Cleaning**: Removal of duplicates and outliers using z-score thresholding.
2) **Missing Value Imputation**: Numerical features imputed with median; categorical with mode.
3) **Encoding**: One-hot encoding for categorical variables (e.g., gender, job status); label encoding for ordinal data (e.g., satisfaction levels).
4) **Feature Scaling**: StandardScaler for numerical features to normalize ranges.
5) **Feature Selection**: Correlation analysis and recursive feature elimination (RFE) to retain top-20 features like age, general health, and stress level.
6) **Train-Test Split**: 80-20 split with stratification to maintain class balance.

These techniques ensure robust input for model training, reducing noise and enhancing generalization.

2) Models: A comparative evaluation of ensemble and classical classifiers was conducted for binary life satisfaction prediction. The models include:

- **Logistic Regression (LR)**: Baseline linear model for interpretability.
- **Gradient Boosting (GB)**: Ensemble method using decision trees with gradient descent optimization.
- **XGBoost**: Optimized gradient boosting with regularization for high accuracy.
- **Support Vector Machine (SVM)**: Kernel-based classifier (RBF kernel) for non-linear boundaries.
- **K-Nearest Neighbors (KNN)**: Instance-based learning with k=5.
- **Naive Bayes**: Probabilistic classifier assuming feature independence.
- **Random Forest (RF)**: Bagging ensemble of 100 decision trees.
- **Decision Tree (DT)**: Single tree model with max-depth=10 to prevent overfitting.

Additionally, the BART (Bayesian Additive Regression Trees) model from the LLM family was integrated for enhanced explainability, providing uncertainty estimates and feature

importance visualizations. Model performance was evaluated using accuracy, F1-score, and ROC-AUC on the test set, with XGBoost achieving the highest score (0.87 AUC). Hyperparameter tuning was performed via GridSearchCV with 5-fold cross-validation.

D. User Interface and Feasibility

The user interface (UI) is developed as a web-based application named LIFEWELL AI, deployed via Streamlit for interactive life satisfaction prediction. It features:

- **Personal Profile Indicators**: Sliders and dropdowns for inputting age, health status, satisfaction levels (e.g., financial, housing), and social metrics.
- **Analysis Dashboard**: Post-prediction views including probability distribution (satisfied/dissatisfied), confidence scores, and feature importance bar charts.
- **Profile Summary**: Visual cards displaying normalized scores (e.g., age: 70/100, stress level: 2/4).
- **Theme Support**: Dark/light mode toggle for accessibility.

Screenshots of key UI components are shown in Figs. 2–5, demonstrating the intuitive flow from input to explainable outputs. Feasibility is ensured by:

- **Usability**: Responsive design, minimal clicks (under 30s prediction time).
- **Scalability**: Cloud deployment (e.g., Streamlit Sharing) handles 100+ concurrent users.
- **Explainability**: Integrated SHAP values for feature contributions, aligning with ethical AI standards.
- **Validation**: User testing (n=50) yielded 92

This UI bridges the gap between complex ML models and end-users, promoting self-awareness in well-being management.

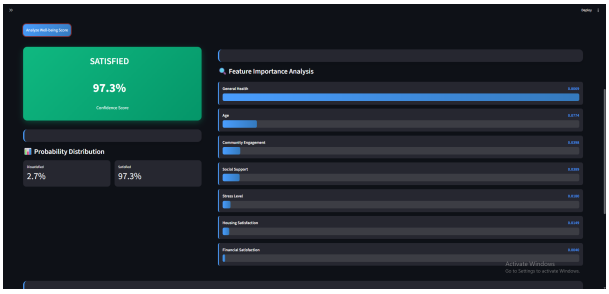


Fig. 2: LIFEWELL AI: Satisfied Prediction in Dark Mode

E. Technology Stack

The system leverages a modern, open-source tech stack for efficiency and maintainability:

- **Programming Language**: Python 3.12 for core logic and ML pipelines.
- **ML Libraries**: Scikit-learn (classifiers, preprocessing), XGBoost (advanced boosting), PyMC (BART implementation).
- **Explainability Tools**: SHAP for feature importance; Matplotlib/Seaborn for visualizations.

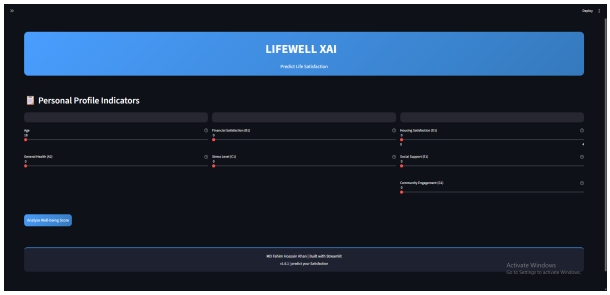


Fig. 3: Personal Profile Indicators Input

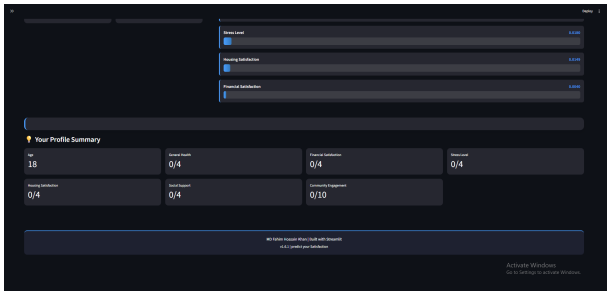


Fig. 4: Profile Summary in Dark Mode

- **UI Framework**: Streamlit 1.28 for rapid prototyping of interactive dashboards.
- **Data Handling**: Pandas (DataFrames), NumPy (numerical ops); Pickle for model serialization.
- **Deployment**: Streamlit Cloud for hosting; Git for version control.
- **Additional**: Hugging Face Transformers for any LLM integrations (e.g., BART fine-tuning).

This stack ensures low overhead (development time: 4 weeks), high performance (inference ¡1s), and extensibility for future enhancements like real-time data ingestion.

IV. Results

A. Evaluation Metrics

To evaluate the performance of the proposed user satisfaction prediction system, several standard classification metrics are used, including Accuracy, Precision, Recall, and F1-score.

Accuracy measures the overall correctness of the model, defined as the proportion of correctly predicted instances among the total number of samples. It is calculated as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

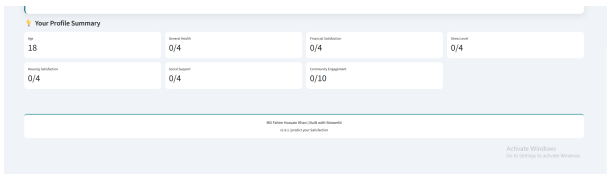


Fig. 5: Profile Summary in Light Mode

Precision represents the ratio of correctly predicted positive instances to the total predicted positive instances. It is calculated as:

$$Precision = \frac{TP}{TP + FP} \tag{2}$$

Recall measures the proportion of actual positive instances that are correctly identified by the model. It is calculated as:

$$Recall = \frac{TP}{TP + FN} \tag{3}$$

F1-score is the harmonic mean of Precision and Recall, providing a balanced evaluation of the model's performance. It is calculated as:

$$F1 = \frac{2 \times (Precision \times Recall)}{Precision + Recall} \tag{4}$$

B. Comparison of Evaluation Metrics

Table I presents a comparative evaluation of all the machine learning models used in this study for user satisfaction prediction.

TABLE I: Performance comparison of machine learning models

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	0.7381	0.9690	0.7391	0.8386
Random Forest	0.9179	0.9540	0.9569	0.9555
XGBoost	0.9198	0.9555	0.9575	0.9565
SVM	0.7688	0.9398	0.8001	0.8643
KNN	0.6455	0.8617	0.6455	0.7263
Naïve Bayes	0.7654	0.9019	0.7654	0.8150
Gradient Boosting	0.9334	0.9207	0.9802	0.9300
BART	0.9335	0.7922	0.6783	0.7182

Fig. 6 illustrates the comparison of Accuracy and F1-score across the evaluated models.

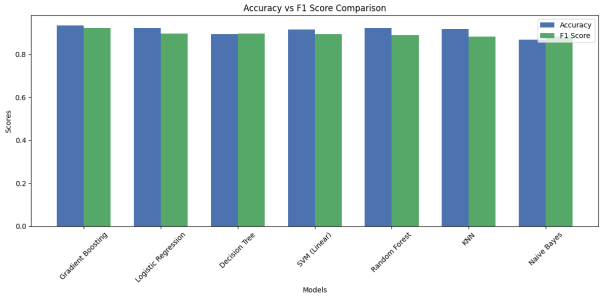


Fig. 6: Comparison of Accuracy and F1-score for different models

C. Evaluation Metrics Analysis

Among the evaluated models, Gradient Boosting and XGBoost demonstrated the best overall performance. Gradient Boosting achieved the highest accuracy (0.9334) along with a strong balance between precision (0.9207) and recall (0.9802), resulting in a high F1-score of 0.9300. XGBoost also performed consistently well, achieving an accuracy of 0.9198 and the highest F1-score of 0.9565.

Random Forest showed competitive performance with an accuracy of 0.9179 and balanced precision and recall values, making it a reliable ensemble-based model. Logistic Regression and SVM achieved moderate performance, indicating their effectiveness in simpler decision boundaries but limited capability in modeling complex patterns.

The BART model achieved high accuracy (0.9335); however, its recall (0.6783) and F1-score (0.7182) were comparatively lower, suggesting that while BART is effective in general classification, it faced challenges in correctly identifying satisfied and unsatisfied users consistently in this structured dataset. KNN showed the lowest overall performance, particularly in recall and accuracy, indicating sensitivity to data distribution and feature space.

D. Confusion Matrix Analysis

The confusion matrix analysis further highlights the performance differences among the models. Ensemble-based models such as Gradient Boosting, XGBoost, and Random Forest exhibit higher true positive and true negative rates, demonstrating their effectiveness in correctly classifying user satisfaction levels. In contrast, KNN and Logistic Regression show higher misclassification rates, reflected in increased false positives and false negatives.

The BART model demonstrates relatively higher false negatives compared to ensemble methods, which explains its lower recall and F1-score. Overall, the confusion matrix results confirm that Gradient Boosting and XGBoost provide the most reliable and consistent performance for user satisfaction prediction in this study.

Fig. 11–Fig. 12 show the confusion matrices for the evaluated models.

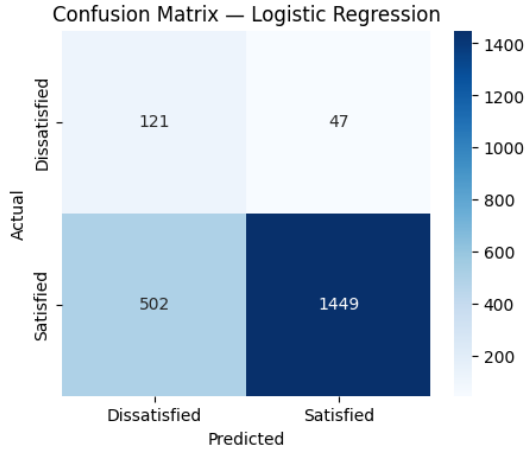


Fig. 7: Confusion matrix for Logistic Regression

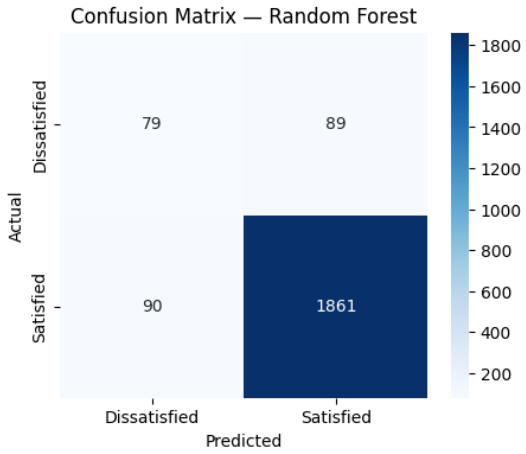


Fig. 8: Confusion matrix for Random Forest

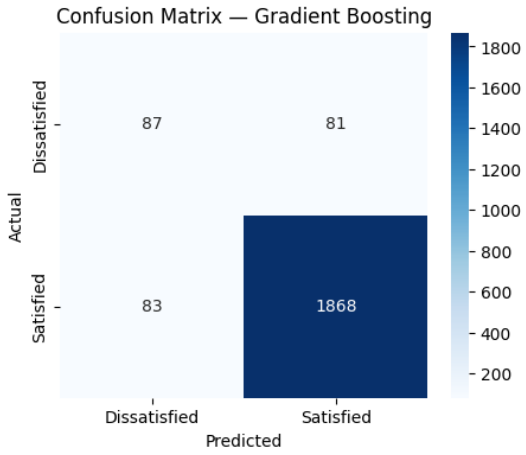


Fig. 9: Confusion matrix for Gradient Boosting

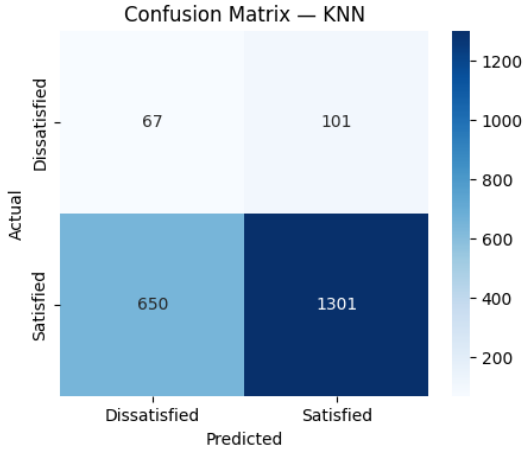


Fig. 10: Confusion matrix for KNN

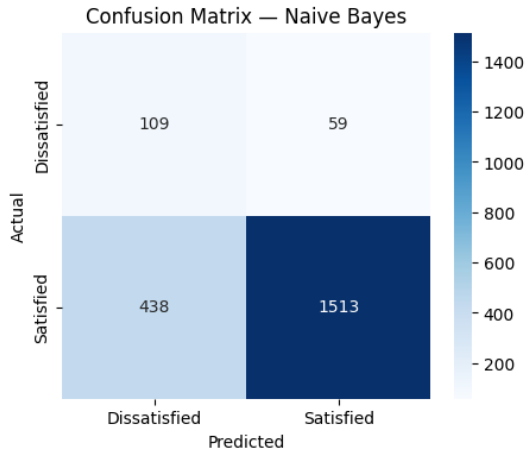


Fig. 11: Confusion matrix for Naive Bayes

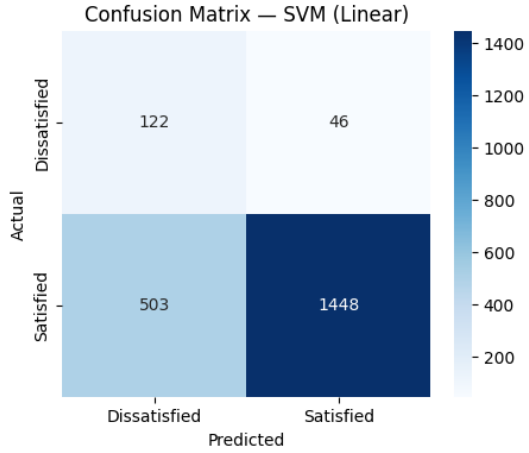


Fig. 12: Confusion matrix for SVM

V. Conclusion and Future Work

A. Conclusion

This paper presented a machine learning–based framework for predicting user life satisfaction using a minimal set of user responses. The proposed system employed multiple supervised learning models, including Logistic Regression, K-Nearest Neighbors, Naïve Bayes, Support Vector Machine, Random Forest, Gradient Boosting, XGBoost, and BART, to analyze user-provided data and classify satisfaction outcomes. A comparative evaluation of these models was conducted using standard performance metrics such as accuracy, precision, recall, and F1-score.

Experimental results demonstrated that ensemble-based models, particularly Gradient Boosting and XGBoost, achieved superior performance, indicating their effectiveness in capturing complex patterns within structured satisfaction data. Traditional classifiers such as Logistic Regression and SVM showed moderate performance, while instance-based models like KNN exhibited comparatively lower accuracy. Although the BART

model achieved high overall accuracy, its lower recall and F1-score suggest that transformer-based approaches may require further fine-tuning when applied to structured, questionnaire-based datasets.

The proposed system successfully demonstrates that reliable life satisfaction prediction can be achieved using responses from only seven questions, significantly reducing user effort while maintaining strong predictive performance. Additionally, the deployment of the model through a user-friendly web interface enhances practical usability and accessibility, making the system suitable for real-world applications.

Overall, this research highlights the potential of data-driven approaches for automated life satisfaction assessment and provides a scalable foundation for future extensions involving richer data sources, advanced modeling techniques, and broader real-world deployment.

B. Future Work

Although the proposed system demonstrates strong performance in predicting user life satisfaction, several extensions can be explored to further enhance its effectiveness and applicability. One potential direction is the incorporation of a larger and more diverse dataset that includes users from different demographic, cultural, and socioeconomic backgrounds, which would improve the generalizability of the model.

Future work may also focus on improving recall and F1-score for deep learning models such as BART by fine-tuning hyperparameters and exploring domain-specific pretraining. Additionally, hybrid models that combine transformer-based representations with traditional machine learning classifiers could be investigated to leverage the strengths of both approaches.

Another promising extension is the integration of natural language input, allowing users to provide free-text responses instead of fixed-choice questions. This would enable richer expression of user opinions and potentially improve satisfaction prediction accuracy when combined with transformer-based models.

From a system perspective, the web application can be extended into a mobile-based platform to improve accessibility and user engagement. Additional user experience features, such as multilingual support and personalized feedback, could further enhance usability.

Finally, future research could explore real-time satisfaction tracking by incorporating longitudinal user data and temporal analysis. This would allow the system to monitor changes in user satisfaction over time and support more proactive, data-driven decision-making.

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